

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

1) The non-oxidative stage of the pentose phosphate pathway _____. 1) _____

- A) produces NADPH and releases CO₂
- B) consumes four ATP molecules
- C) contains two reactions whose enzymes are allosterically inhibited by NADPH
- D) consists entirely of near-equilibrium reactions**

Answer: D

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2) Glucose is stored as _____ in bacteria and animals. 2) _____

- A) glycogen**
- B) NADPH
- C) starch
- D) glucagons

Answer: A

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3) An intermediate found in gluconeogenesis and not glycolysis is 3) _____

- A) fructose 1,6-*bis*phosphate.
- B) oxaloacetate.**
- C) phosphoenolpyruvate.
- D) 2-phosphoglycerate.

Answer: B

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4) In the Cori cycle, gluconeogenesis occurs in _____ and glycolysis in _____. 4) _____

- A) liver; muscle**
- B) liver; liver
- C) muscle; liver
- D) muscle; muscle

Answer: A

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5) Glycogen degradation occurs in 5) _____

- A) saliva.
- B) muscle.
- C) brain cells.
- D) liver.
- E) muscle and liver.**

Answer: E

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- 6) Gluconeogenesis uses the same enzymatic reactions of glycolysis except for the _____. 6) _____
- A) 4 irreversible reactions in glycolysis B) 2 irreversible reactions in glycolysis
- C) 3 irreversible reactions in glycolysis** D) pyruvate kinase catalyzed step

Answer: C

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- 7) Which of the following is not a precursor for gluconeogenesis? 7) _____
- A) glycerol
- B) oxaloacetate
- C) alanine
- D) acetate
- E) ethanol**

Answer: E

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- 8) Glycerol is converted to _____ when it is used for gluconeogenesis. 8) _____
- A) 3-phosphoglycerate **B) dihydroxyacetone phosphate**
- C) phosphoenolpyruvate D) oxaloacetate

Answer: B

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TRUE/FALSE. Write 'T' if the statement is true and 'F' if the statement is false.

- 9) UDP-glucose is a precursor of glycogen in all cells. 9) _____

Answer: FALSE

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- 10) Two molecules of NADPH are generated for each molecule of glucose 6-phosphate that enters the pentose phosphate pathway. 10) _____

Answer: **TRUE**

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